Fabric & Clothing Store POS

Internship assignment report: setup, design, test results and known limitations

1. What was built

A local Windows-friendly POS web application (Node.js 22 + built-in SQLite, vanilla JavaScript UI, no external dependencies) for a fabric and clothing store. Quantities are stored as integer thousandths and money as integer paisa, so there is no floating-point rounding in totals. All business rules run on the server inside database transactions; the browser never decides totals.

Stack note: the brief suggests React + Vite + Express + Electron. This build uses a dependency-free Node server and a plain JavaScript UI so it runs offline with one command. The stack should be confirmed with the supervisor; an Electron wrapper (electron/main.js) is included but was **not built or tested**, and no installer was produced.

2. How to run (Windows PowerShell)

Requires Node.js 22 or newer. In the project folder: `npm run seed` (optional demo data, every item marked [DEMO]), then `npm start`, then open http://localhost:3000. For a clean database skip the seed (or delete data/pos.db); a first-run admin account is created automatically. Tests: `npm test`. Restore a backup: stop the server and run `node scripts/restore.js data/backup-XXXX.db`.

Role	Login (demo)	Allowed
admin	admin / admin123	Everything, including settings, users, backup
manager	manager / manager123	Sales, returns/exchanges, purchases, products, reports, expenses, stock adjustment, audit log, stock override approval
cashier	cashier / cashier123	Sales, customers, shift open/close only

Change the default passwords before real use.

3. Design decisions

Topic	Decision
Costing method	Weighted average. Each purchase recalculates the product cost; every sale line stores a cost snapshot used by profit reports.
Profit	Gross profit estimate = net sale value (after discounts and returns) minus cost of goods sold. Operating expenses are **not** deducted; they are shown separately. Revenue is never labelled profit.
Units / precision	Each product has its own unit (m, yd, pc, suit, roll, kg). Length units allow up to 3 decimals; piece/suit units reject fractions. Prices are per the product unit.
Suits	A suit can be one SKU or linked components; selling it deducts each component exactly once; its cost is the sum of component costs.
Rolls	Rolls are created when purchases are received with a Roll ID; a cashier may pick a roll at checkout and its remaining length is reduced.
Returns / exchange	Original invoice is never edited; returns and exchanges create new records, adjust stock (if restocked), and reduce the customer balance. Exchange (return + new sale) is one atomic transaction.
Negative stock	Blocked. A manager/admin can authorize a logged override by entering their own credentials at the point of sale.
Data safety	Checkout, return, exchange and purchase receiving each run in one transaction. A one-time token per cart stops double-submit. Reprint reads the saved invoice only.
Printing	Three separate templates: 58mm and 80mm thermal (narrow, monospace) and A4 (table, logo, signatures). Labels support 40x25, 50x30 and 58x40 mm. Every print opens a preview window first. Barcode symbology is Code 39.

4. Acceptance test checklist (section 10 of the brief)

Test	Expected	Result	Evidence
Sell 2.35 m at PKR 850/m	PKR 1,997.50; stock -2.35 m	PASS	Automated test + browser run (receipt shows 1,997.50)
Sell 2 pieces at PKR 1,200	PKR 2,400; stock -2	PASS	Automated test + browser run
Component-based suit	Components deducted once	PASS	Automated test; browser: 1001 and 1002 reduced by 2.5 m and 2.0 m
Insufficient meter stock	Blocked or logged override	PASS	Automated: blocked, nothing saved; override requires manager login and is logged
Partial payment / udhaar	Correct balance and ledger	PASS	Automated + browser (PKR 950 sale, 100 paid, balance 850)
Return from multi-item invoice	Only returned qty reversed	PASS	Automated + browser (0.5 m returned, refund offset against balance)
Reprint invoice	No new sale, no stock change	PASS	Reprint reads saved invoice; invoice count unchanged after 58/80/A4 prints
58mm / 80mm / A4 print	Correct template, totals match	PASS (preview only)	Print-preview screenshots below. **Not tested on physical printers.**
Scan barcode	Correct item; bad/duplicate handled	PASS	Browser run typed barcodes into scan field (keyboard-wedge behaviour); unknown barcode shows an error; duplicate barcode rejected by database
Restart application	Data persists	PASS	Data file reopened in a separate process; all sales, payments and returns present
Reports	Date-filtered totals match	PARTIAL	Dashboard net sales and gross profit verified by automated test; other report types were not independently reconciled

Automated tests: **tests 18 | pass 18 | fail 0 |** (meter, piece, suit, stock block and override, partial payment, return, exchange, cash-refund ledger, purchase receiving, purchase return, shift cash, stock adjustment, variants, tax and invoice maths, login, role permissions, duplicate submit).

5. Requirement coverage

Area	Status
Dashboard (today default, date filters, low stock, top items, payment split, profit)	Done, database-driven
POS: barcode, search, category filter, fractional qty, line/bill discount, tax, customer, hold/resume, split payment, change/balance, roll selection	Done
Products & variants (all listed fields, size variants with own barcode/stock, auto barcode, activate/deactivate, labels)	Done
Roll inventory and stock movement history, adjustments, damage	Done; reserved length not supported
Suits/sets with components	Done
Purchases, suppliers, supplier balance and payments, purchase returns, expenses	Done
Customers, ledger, credit limit, ledger payments	Done
Returns, exchange (refund or balance offset), permission control	Done
Reports (product, category, employee, payment method, roll balances, customer balances, returns, expenses, purchases, supplier balances), print and CSV export	Done
Users, roles, login, shifts, audit log, protected settings	Done
Settings: store info, logo, tax, invoice prefix, receipt size, backup, restore (script), export	Done; see limitations

Area	Status
Electron desktop build / installer	Wrapper included, **not built or tested**
Demo video	Not included

6. Known limitations

• Not tested on physical 58mm/80mm/A4 printers; the printer is chosen in the Windows print dialog and the paper size comes from the template.

• Currency code is stored as a setting but prices on screen and receipts are always shown as PKR.

• Wholesale price is stored but not applied automatically at the POS; unit conversion (m to yd) is not implemented.

• Barcodes use Code 39; an optional QR code on receipts is not included. Roll-level reserved length is not supported.

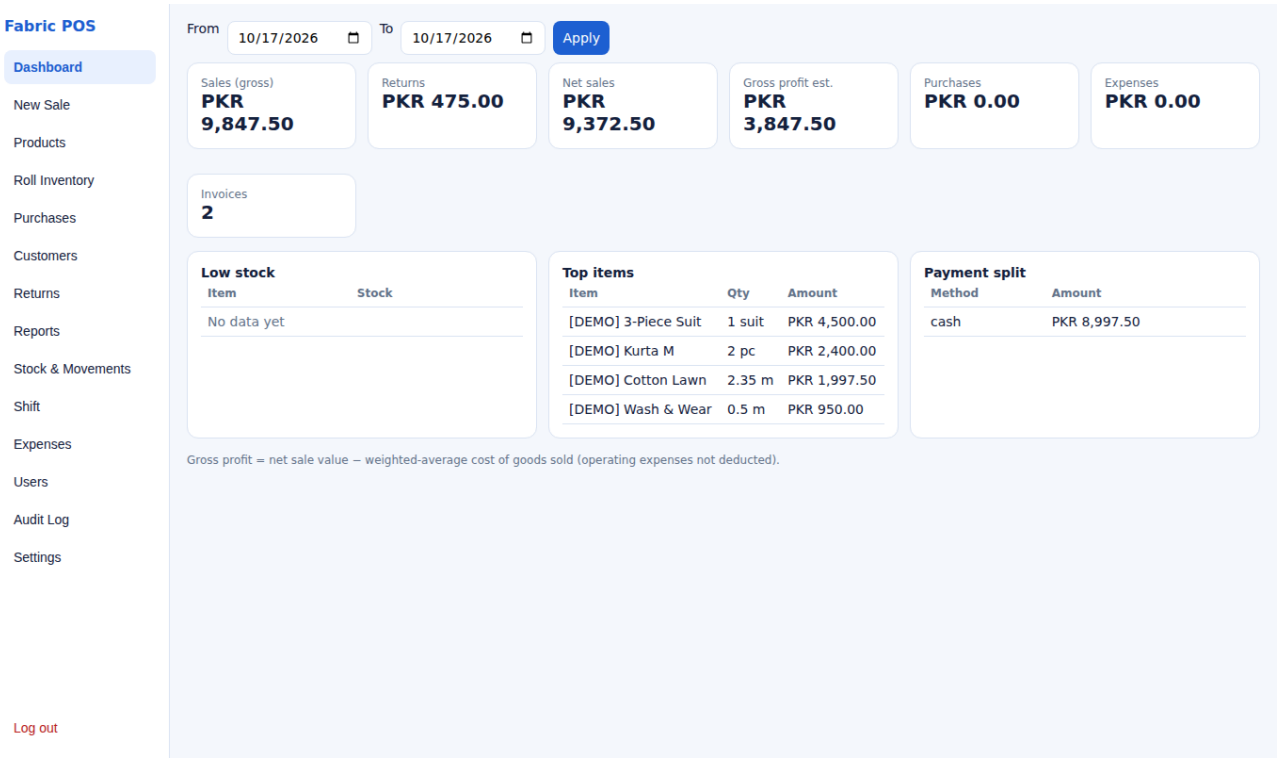
• Restore requires stopping the server and running a script (no in-app restore button).

• Sessions are held in server memory (logging in again is needed after a server restart). Built for a single-store local machine, not for internet exposure.

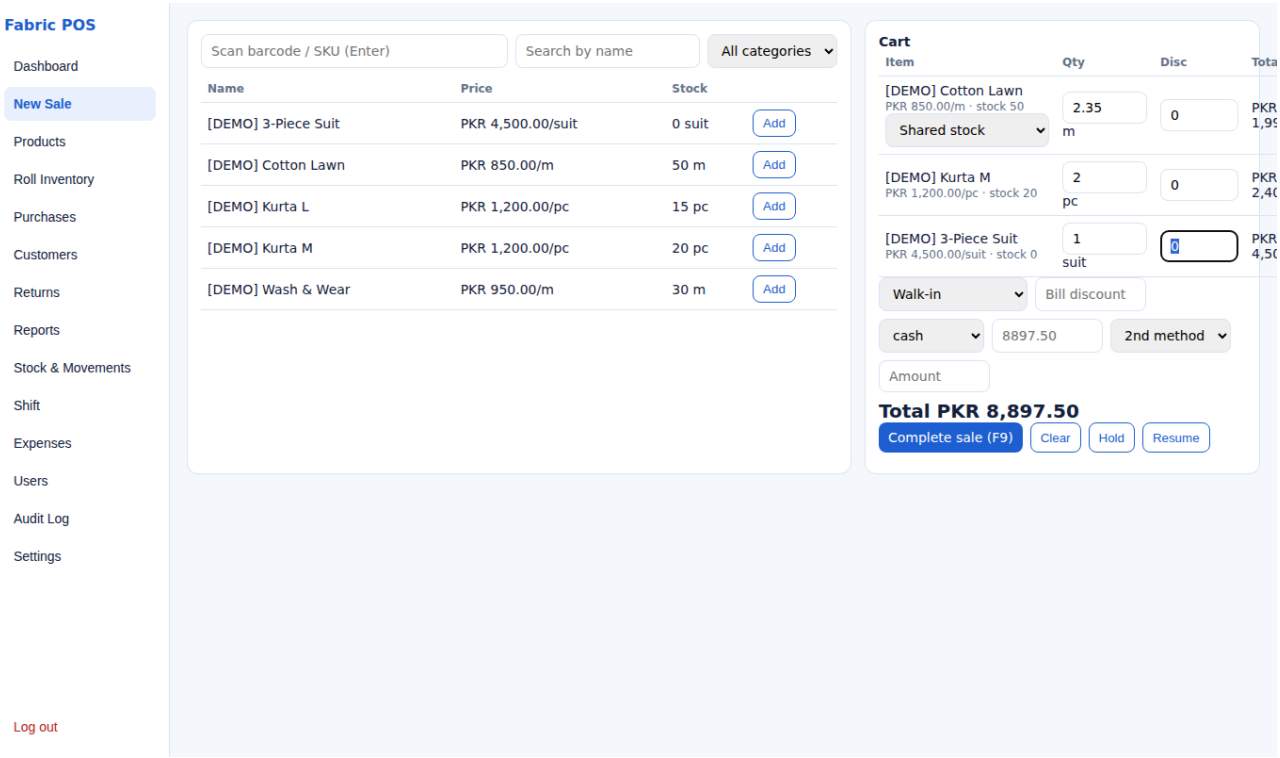
• The UI is plain JavaScript, not React/Vite; no Electron installer was produced.

7. Screenshots (captured from the running app)

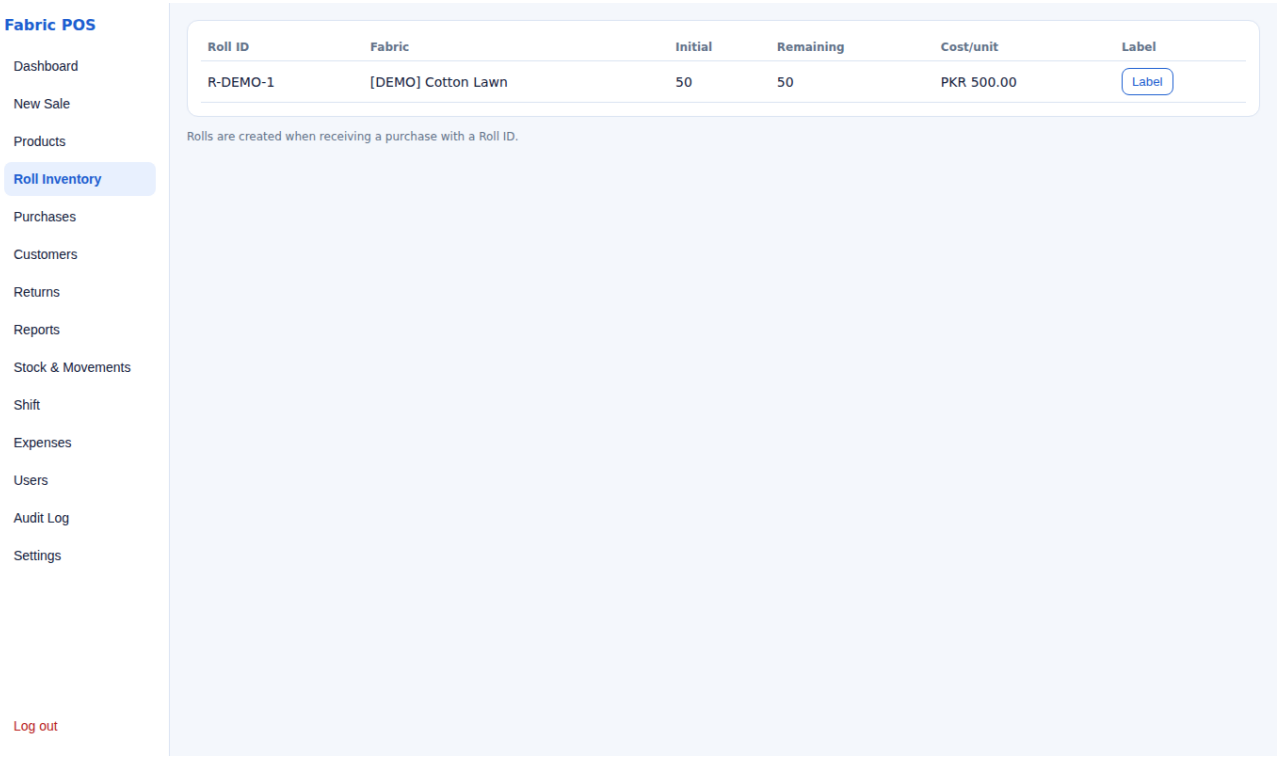
Dashboard



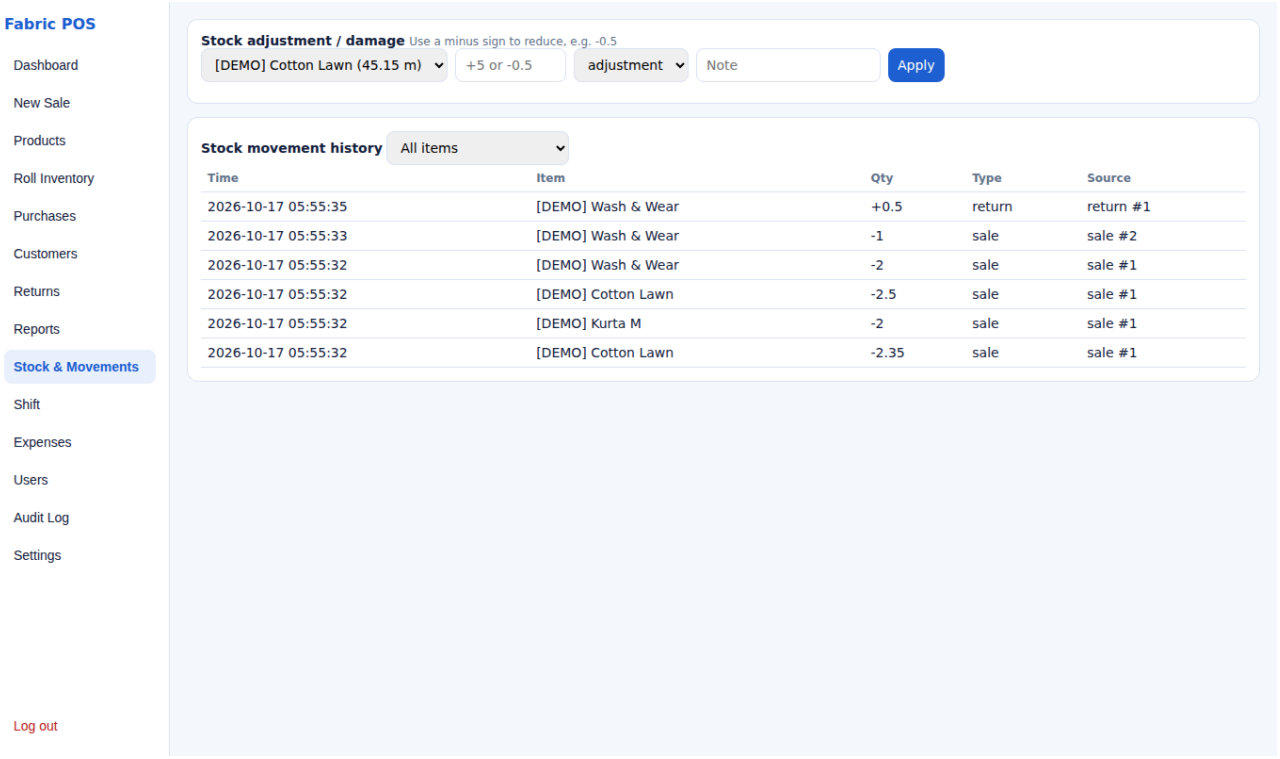
POS checkout (meter, piece and suit lines)



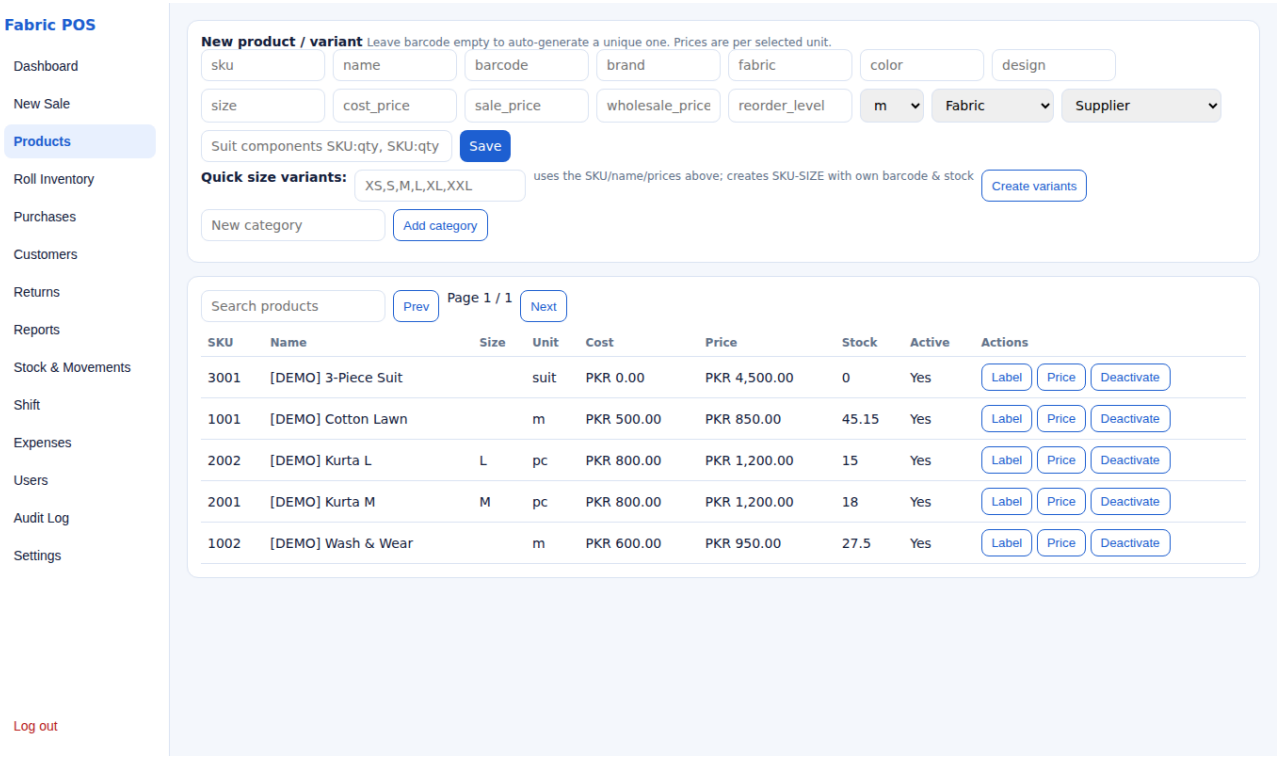
Roll inventory



Stock movements



Products & variants



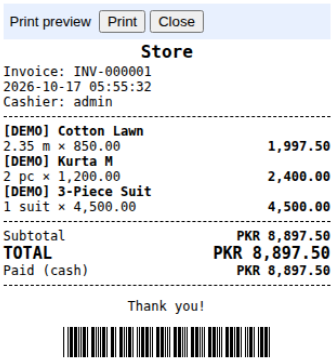
Barcode label (roll)



58mm receipt preview



80mm receipt preview



A4 invoice preview

Print preview Print Close

Store

Invoice: INV-000001
2026-10-17 05:55:32
Cashier: admin

#	Item	Qty	Price	Disc	Total
1	[DEMO] Cotton Lawn	2.35 m	PKR 850.00	PKR 0.00	PKR 1,997.50
2	[DEMO] Kurta M	2 pc	PKR 1,200.00	PKR 0.00	PKR 2,400.00
3	[DEMO] 3-Piece Suit	1 suit	PKR 4,500.00	PKR 0.00	PKR 4,500.00

Subtotal	**PKR 8,897.50**
TOTAL	**PKR 8,897.50**
Paid (cash)	**PKR 8,897.50**

Payment status: **PAID**

Customer signature _____ Authorized signature _____

Thank you!